Standard Operating Procedure

Cable Labeling & Termination Standards

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Related Documents	SOP-LV-001, SOP-LV-013, REF-AC-RS485-001
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Applies To	Field Technicians, Installers
Jurisdiction	Ontario / Manitoba
Owner	Apex – Operations
Approved By	______________________________

1. Purpose

This procedure establishes a consistent cable labeling and termination standard across all Apex low-voltage work — structured cabling, access control, and camera systems alike — so that any technician can trace, identify, and service a cable run without relying on the original installer's memory.

2. Scope

Applies to all copper and fiber cabling installed by Apex technicians, including Cat6/Cat6A structured cabling, RS-485 access control runs, Wiegand reader runs, and camera cabling.

3. Labeling Standard — Based on ANSI/TIA-606

Apex labeling practice follows the principles of ANSI/TIA-606 (the telecommunications infrastructure administration standard), adapted for field use:

- Every cable is labeled at both ends — never only one end. A cable labeled at only one end is effectively unlabeled the next time someone is troubleshooting from the far side.
- Labels are placed within approximately 300 mm (12 in) of each termination point — close enough to the connection that there's no ambiguity about which cable the label belongs to.
- Labels must be durable and legible for the life of the installation — use a proper label printer/wrap-around label, not handwritten tape or permanent marker directly on the jacket, which fades and smudges over time.
- Use a consistent identifier scheme across a site — e.g., room/location + device type + sequence number — matching the site's as-built documentation (see REF-AC-RS485-001 for the wiring reference format used on access control runs).
- Label patch panels, punch-down blocks, racks, and enclosures, not just individual cables — a fully labeled cable run means nothing if the panel it lands on is unmarked.

4. Cat6 / Cat6A Termination Standard

- Use a single wiring standard (T568B, unless the site's existing infrastructure is already standardized on T568A — confirm before starting rather than mixing standards on one site) consistently across the entire job.
- Maintain twist as close to the termination point as possible — industry guidance is typically no more than 1/2 inch (13 mm) of untwisted conductor at the termination, since excessive untwist degrades crosstalk performance.
- Do not exceed the cable's minimum bend radius (typically 4x the cable's outer diameter for most Cat6/6A — confirm against the specific cable's datasheet) when routing around corners or through pathways.
- Dress and secure cable runs neatly in pathways/trays — avoid excessive cinching with zip ties, which can deform the cable and affect performance; use velcro or properly spaced, hand-tight ties instead.

5. RS-485 & Low-Voltage Control Wiring Termination

- Strip insulation to the minimum length needed for the terminal — excess bare conductor risks shorting to an adjacent terminal; insufficient strip length risks a poor connection.

- Avoid over-tightening terminal screws, which can nick or shear conductor strands — see SOP-LV-013 for RS-485-specific termination and bias requirements.
- Label control wiring (REX, DPS, lock outputs, RS-485) at the controller terminal block and at the device end, using the same identifier scheme as structured cabling — not an ad hoc shorthand that only makes sense to the technician who installed it.

6. Documentation

Every labeled run should appear in the site's as-built documentation — a physical label that doesn't match any record, or a record that doesn't match any physical label, both defeat the purpose of labeling. Update site wiring references (REF-AC-RS485-001 format, or the site's structured cabling as-built) as work is completed, not from memory afterward.

7. References

- ANSI/TIA-606-D — Administration Standard for Telecommunications Infrastructure
- ANSI/TIA-568 series — Commercial Building Telecommunications Cabling Standard
- SOP-LV-001 — Drop Ceiling Cat6 / Low-Voltage Installation
- SOP-LV-013 — RS-485 Bus Design & Installation
- REF-AC-RS485-001 — site wiring reference format